Highlights

A Cell-Centered Lagrangian-AMR Scheme Based on a Kinematically Constrained Hanging Nodal Solver

Chunyuan Xu

- A cell-centered Lagrangian-AMR scheme is developed using a decoupled, kinematically constrained hanging nodal solver.
- The constraint force is computed from the nodal momentum balance after enforcing the kinematic constraint.
- Semi-discrete conservation of mass, momentum, and energy is mathematically proven, while the Geometric Conservation Law (GCL) is also satisfied.
- Grid convergence studies show that non-conforming meshes achieve error levels comparable to those of uniform meshes, while total energy conservation is confirmed across all test cases.

A Cell-Centered Lagrangian-AMR Scheme Based on a Kinematically Constrained Hanging Nodal Solver

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ABSTRACT

Ensuring conservation of mass, momentum, and energy across non-conforming interfaces remains a challenge for cell-centered Lagrangian hydrodynamics (CCH) on locally refined grids. In this paper, we construct a cell-centered Lagrangian-AMR scheme based on a dedicated hanging nodal solver for non-conforming quadrilateral meshes. The central element of this scheme is a decoupled formulation: the hanging node is kinematically constrained to its adjacent master nodes via a collinearity constraint, and the constraint force required to maintain this condition is computed from the nodal momentum balance. The constraint force is incorporated into the governing equations through an internal-energy-weighted distribution, yielding a robust and conservative numerical framework that satisfies the Geometric Conservation Law (GCL) during mesh deformation.

We validate the scheme on both fixed-topology non-conforming meshes and Adaptive Mesh Refinement (AMR) frameworks. A Taylor–Green vortex convergence study demonstrates that non-conforming meshes achieve error levels and convergence rates comparable to those of uniform meshes, while Sod and Sedov tests confirm correct wave transmission across refinement interfaces. Furthermore, integration with AMR managed by the `p4est` library demonstrates that the solver maintains total energy conservation under frequent grid adaptation while remaining geometrically robust under severe grid deformation.

1. Introduction

Numerical simulations of multi-scale compressible flows—such as inertial confinement fusion (ICF) and astrophysical explosions—are characterized by intense shock waves and complex material interfaces. Lagrangian methods are widely used for these problems because they naturally track material interfaces without numerical mixing and automatically focus grid resolution in compression zones [6, 12, 15]. However, the dynamic range of spatial and temporal scales in such flows often exceeds the resolving capacity of uniform meshes at a reasonable computational cost. Local mesh refinement is therefore essential to provide high resolution in critical regions while keeping the overall problem size manageable. Extending the classical concepts of adaptive mesh refinement (AMR) from Eulerian grid frameworks [3] to the Lagrangian setting has therefore attracted considerable interest.

The development of Lagrangian and Arbitrary Lagrangian-Eulerian (ALE) AMR frameworks has progressed from early patch-based architectures [1, 7, 10, 14] toward more localized, cell-by-cell refinement strategies [13]. The majority of these AMR developments have been established within the traditional Staggered-Grid Hydrodynamics (SGH) framework, which places thermodynamic variables at cell centers and velocities at grid nodes. In recent years, Cell-Centered Lagrangian Hydrodynamics (CCH) schemes [11, 12] have emerged as a prominent alternative. Benefiting from a fully conservative formulation, CCH schemes define both thermodynamic and kinetic states within the cells and determine node velocities using a multidimensional Riemann solver (nodal solver). Recently, Colaitis et al. [7] extended this framework to cell-by-cell AMR through a cell-centered AMR-ALE framework for multi-material hydrodynamics. While powerful, such frameworks increase geometric complexity and must carefully preserve volume conservation during mesh motion. To circumvent these complications, we pursue a different avenue by adopting cell-by-cell refinement strictly on quadtree meshes. Rather than allowing cells to distort into arbitrary shapes, we exploit the flexibility of the CCH nodal solver to naturally embed kinematic constraints directly at the hanging nodes, thereby maintaining both geometric simplicity and semi-discrete conservation.

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A central challenge in these localized refinement approaches is the treatment of non-conforming interfaces, where hanging nodes must be constrained to maintain geometric compatibility without violating physical conservation. In classical computational mechanics, non-conforming interface compatibility is typically enforced using Lagrange-multiplier-based mortar element methods [4] or penalty-based formulations [2, 16]. Although standard finite element methods typically enforce field continuity across non-matching grid interfaces during matrix assembly, extending this approach directly to cell-centered Lagrangian hydrodynamics remains challenging. The Lagrangian movement of the mesh requires that the interface treatment simultaneously: (i) satisfy the geometric conservation law (GCL), (ii) conserve mass, momentum, and total energy, and (iii) remain compatible with the Riemann solver. Standard penalty methods generally do not guarantee exact discrete conservation unless specifically symmetrized or embedded into a conservative variational formulation, whereas fully coupled Lagrange multiplier formulations require solving expensive global linear systems.

To address these challenges, we construct a robust cell-centered Lagrangian-AMR scheme based on a hanging nodal solver on quadtree meshes. Recognizing that the unconstrained nodal solver of Maire et al. [11, 12] can be mathematically interpreted as the minimizer of a quadratic velocity-based functional, we formulate the midpoint collinearity of the hanging node as a kinematic constraint on this functional. Rather than solving a coupled system for all nodes, which would entail substantial algorithmic complexity, we propose a decoupled strategy. The master-node velocities are computed independently using the baseline nodal solver, and the hanging-node velocity is then kinematically constrained. The generalized constraint force (Lagrange multiplier) required to enforce this collinearity is computed *a posteriori* from the nodal momentum balance. To close the governing equations conservatively, this constraint force is distributed back to the adjacent cells using an internal-energy-weighted partition, which ensures global conservation of momentum and energy. Compared qualitatively with alternative non-conforming strategies, this decoupled approach preserves the original quadtree topology without requiring complex arbitrary-polygon tracking, and avoids tuning empirical penalty parameters.

The remainder of this paper is organized as follows. Section 2 reviews the governing equations and the baseline cell-centered Lagrangian scheme. Section 3 details the core of our AMR scheme: a decoupled kinematically constrained projection formulation derived from the variational structure of the nodal solver. This incorporates a thermodynamic force redistribution scheme, accompanied by proofs of conservation and GCL compliance. Section 4 validates the solver's numerical properties through studies on fixed-topology non-conforming meshes to isolate the interface treatment. Section 5 extends the scheme to AMR with first-order conservative remapping, with a clear discussion of its performance. Finally, Section 6 summarizes the key findings and outlines future research.

2. Governing Equations and Baseline Cell-Centered Lagrangian Scheme

2.1. Governing equations

The two-dimensional Lagrangian equations of gas dynamics for an inviscid, compressible fluid describe the conservation of volume, mass, momentum, and energy in a moving frame. For a control volume $\Omega(t)$ bounded by its closed boundary $\partial\Omega(t)$ with unit outward normal $\boldsymbol{N}$, the integral conservation laws are:

$$\frac{d}{dt}\int_{\Omega} d\Omega - \int_{\partial\Omega} \boldsymbol{U}\cdot\boldsymbol{N}\, dL = 0, \tag{1a}$$

$$\frac{d}{dt}\int_{\Omega} \rho\, d\Omega = 0, \tag{1b}$$

$$\frac{d}{dt}\int_{\Omega} \rho\boldsymbol{U}\, d\Omega + \int_{\partial\Omega} P\boldsymbol{N}\, dL = 0, \tag{1c}$$

$$\frac{d}{dt}\int_{\Omega} \rho E\, d\Omega + \int_{\partial\Omega} P\boldsymbol{U}\cdot\boldsymbol{N}\, dL = 0. \tag{1d}$$

where t represents time, ρ is the fluid density, P is the pressure, $\boldsymbol{U}$ is the velocity vector, E is the specific total energy, and dL is the length element along the boundary. Equation (1a) represents the Geometric Conservation Law (GCL). The system is closed by a thermodynamic equation of state (EOS): $P = P(\rho, e)$, where the specific internal energy is $e = E - \frac{1}{2}|\boldsymbol{U}|^2$. For an ideal gas, the pressure is given by

$$P = (\gamma - 1)\rho e, \tag{2}$$

where γ denotes the adiabatic index.

2.2. The cell-centered Lagrangian scheme on conforming quadrilateral meshes

Our hanging nodal solver builds on the cell-centered Lagrangian hydrodynamics (CCH) scheme of Maire et al. [12]. This section reviews the baseline discretization on conforming quadrilateral meshes, establishing the notation and the conservation framework into which the hanging-node treatment will be embedded.

2.2.1. Notation and assumptions

The two-dimensional spatial domain is discretized into quadrilateral cells. Each cell, denoted by Ω_c, is assigned a unique index c, and each vertex is assigned a unique index p. As shown in Fig. 1, for a given cell Ω_c and a node $p \in \mathcal{P}(c)$ (where $\mathcal{P}(c)$ is the set of vertices of cell c), the two neighboring vertices along the cell boundary are denoted by p^+ and p^- (in the counterclockwise and clockwise directions, respectively).

- **Fluid Variables and Lagrangian Mass:** The thermodynamic and dynamic state of cell c is defined by the cell-centered density ρ_c, pressure P_c, specific total energy E_c, and velocity vector $\boldsymbol{U}_c$. The cell mass $m_c = \rho_c(t)V_c(t)$ is invariant in time, where $V_c(t)$ is the cell volume. For any vertex $p \in \mathcal{P}(c)$, its kinematic state is specified by position $\boldsymbol{X}_p(t)$ and velocity $\boldsymbol{U}_p$.

- **Half-Edge Notation:** Each cell boundary edge at node p is split into half-segments. Let $L_{pp^\pm}$ and $\boldsymbol{N}^c_{pp^\pm}$ be the total lengths and unit outward normals of the adjacent edges with respect to cell c. The subcell half-edge lengths and unit outward normals are defined as:

$$L^c_{\overline{p}} = \frac{1}{2} L_{pp^+}, \quad \boldsymbol{N}^c_{\overline{p}} = \boldsymbol{N}^c_{pp^+}, \qquad L^c_{\underline{p}} = \frac{1}{2} L_{pp^-}, \quad \boldsymbol{N}^c_{\underline{p}} = \boldsymbol{N}^c_{pp^-}. \tag{3}$$

  The subcell half-edge pressures $\Pi^c_{\overline{p}}$ and $\Pi^c_{\underline{p}}$ act as auxiliary thermodynamic variables at the node, which are used to construct the numerical fluxes.

Under these definitions, the rate of change of the cell volume V_c is directly linked to the nodal velocities $\boldsymbol{U}_p$ [11], enabling a discretization framework that satisfies the GCL.

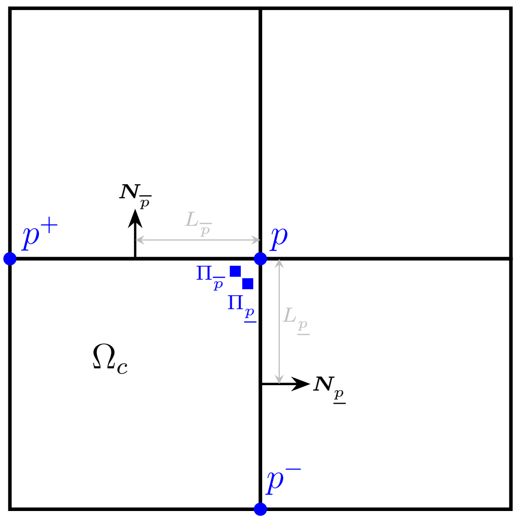


Figure 1: Schematic of the notation for a generic quadrilateral cell Ω_c, showing the subcell parameters $(L^c_{\overline{p}}, \boldsymbol{N}^c_{\overline{p}}, \Pi^c_{\overline{p}})$ and $(L^c_{\underline{p}}, \boldsymbol{N}^c_{\underline{p}}, \Pi^c_{\underline{p}})$ associated with node p and its neighbors p^+ and p^-.

2.2.2. *Conservation-based nodal solver*

Following Maire et al. [12], a multidimensional acoustic Riemann solver is formulated at each node. For a generic cell Ω_c sharing node p, the subcell half-edge pressures are linearized using a one-dimensional acoustic approximation:

$$\begin{aligned} \Pi^c_{\overline{p}} &= P_c + Z_c(\boldsymbol{U}_p - \boldsymbol{U}_c) \cdot \boldsymbol{N}^c_{\overline{p}}, \\ \Pi^c_{\underline{p}} &= P_c + Z_c(\boldsymbol{U}_p - \boldsymbol{U}_c) \cdot \boldsymbol{N}^c_{\underline{p}}, \end{aligned} \tag{4}$$

where $Z_c = \rho_c a_c$ is the local acoustic impedance, with a_c being the sound speed. To ensure momentum conservation of the cell-centered scheme, the subcell forces at node p must sum to zero (force balance at the node):

$$\sum_{c \in \mathcal{C}(p)} \left(\Pi^c_{\overline{p}} L^c_{\overline{p}} \boldsymbol{N}^c_{\overline{p}} + \Pi^c_{\underline{p}} L^c_{\underline{p}} \boldsymbol{N}^c_{\underline{p}} \right) = \boldsymbol{0}, \tag{5}$$

where $\mathcal{C}(p)$ is the set of cells connected to vertex p. Substituting Eq. (4) into Eq. (5) yields the explicit nodal velocity:

$$\boldsymbol{U}_p = \mathsf{M}_p^{-1} \sum_{c \in \mathcal{C}(p)} \left(\mathsf{M}^c_p \boldsymbol{U}_c - P_c L^c_p \boldsymbol{N}^c_p \right), \tag{6}$$

where the corner normal vector $L^c_p \boldsymbol{N}^c_p$, the subcell matrix M^c_p, and the nodal matrix M_p are defined as:

$$L^c_p \boldsymbol{N}^c_p = L^c_{\overline{p}} \boldsymbol{N}^c_{\overline{p}} + L^c_{\underline{p}} \boldsymbol{N}^c_{\underline{p}}, \quad \mathsf{M}_p = \sum_{c \in \mathcal{C}(p)} \mathsf{M}^c_p, \quad \mathsf{M}^c_p = Z_c \left(L^c_{\overline{p}} \boldsymbol{N}^c_{\overline{p}} \otimes \boldsymbol{N}^c_{\overline{p}} + L^c_{\underline{p}} \boldsymbol{N}^c_{\underline{p}} \otimes \boldsymbol{N}^c_{\underline{p}} \right), \tag{7}$$

with $\otimes$ denoting the tensor product. Substituting $\boldsymbol{U}_p$ back into Eq. (4) closes the nodal solver.

Remark on the minimization structure. The nodal velocity $\boldsymbol{U}_p$ in Eq. (6) can also be derived as the unique minimizer of the quadratic functional:

$$\mathcal{J}(\boldsymbol{U}_p) = \frac{1}{2} \boldsymbol{U}_p \cdot \left(\mathsf{M}_p \boldsymbol{U}_p \right) - \boldsymbol{U}_p \cdot \sum_{c \in \mathcal{C}(p)} \left(\mathsf{M}^c_p \boldsymbol{U}_c - P_c L^c_p \boldsymbol{N}^c_p \right). \tag{8}$$

Provided that the surrounding subcell normals span $\mathbb{R}^2$, the matrix M_p is symmetric positive definite and the minimizer is unique. Setting $\partial \mathcal{J} / \partial \boldsymbol{U}_p = \boldsymbol{0}$ recovers Eq. (6). This least-squares structure provides the foundation for the constrained formulation developed in Section 3.

2.3. Spatial discretization on conforming meshes

The semi-discrete finite volume equations are obtained by integrating the conservation laws over the moving cell $\Omega_c(t)$. For cell c, the subcell force exerted at node p is:

$$\boldsymbol{F}_{cp} = \Pi^c_{\overline{p}} L^c_{\overline{p}} \boldsymbol{N}^c_{\overline{p}} + \Pi^c_{\underline{p}} L^c_{\underline{p}} \boldsymbol{N}^c_{\underline{p}}. \tag{9}$$

Since the cell mass m_c is constant, the semi-discrete conservation equations for cell c read:

$$m_c \frac{d}{dt}\left(\frac{1}{\rho_c} \right) - \sum_{p \in \mathcal{P}(c)} L^c_p \boldsymbol{N}^c_p \cdot \boldsymbol{U}_p = 0, \tag{10a}$$

$$m_c \frac{d\boldsymbol{U}_c}{dt} + \sum_{p \in \mathcal{P}(c)} \boldsymbol{F}_{cp} = \boldsymbol{0}, \tag{10b}$$

$$m_c \frac{dE_c}{dt} + \sum_{p \in \mathcal{P}(c)} \boldsymbol{F}_{cp} \cdot \boldsymbol{U}_p = 0. \tag{10c}$$

where Eq. (10a) represents the GCL, while Eqs. (10b) and (10c) represent the conservation of momentum and total energy, respectively. The mesh vertex coordinates evolve according to the kinematic equation:

$$\frac{d}{dt} \boldsymbol{X}_p(t) = \boldsymbol{U}_p(t), \quad \boldsymbol{X}_p(0) = \boldsymbol{x}_p. \tag{11}$$

Equations (10)–(11) form a fully conservative system on conforming grids and serve as the baseline for the non-conforming extension.

3. Decoupled Kinematically Constrained Hanging Nodal Solver

Extending the CCH scheme to meshes with non-conforming interfaces requires treating numerical fluxes at hanging nodes. This section develops the central contribution of this paper: a decoupled kinematically constrained formulation that enforces collinearity, computes the constraint force *a posteriori*, and preserves all conservation properties.

3.1. Notation and assumptions on non-conforming meshes

We consider fixed-topology non-conforming quadrilateral meshes in which refinement regions are prescribed *a priori* and the grid topology remains fixed throughout the simulation (dynamic adaptation is discussed in Section 5). As illustrated in Fig. 2, a typical non-conforming interface involves a coarse parent cell Ω_{parent} adjacent to two fine child cells. We distinguish two types of nodes:

- **Master nodes (or regular nodes)**, denoted by $p \in \mathcal{P}$, are standard mesh vertices whose motion is computed by the standard nodal solver (6).
- **Hanging nodes**, denoted by $h \in \mathcal{H}$, lie on the refinement boundary at the midpoint of the segment connecting two master nodes p_1 and p_2. Their motion is kinematically constrained.

No hanging nodes arise on physical boundaries of the computational domain. When cells on a physical boundary are refined, the newly introduced boundary nodes do not interface with a coarser adjacent cell. Consequently, they are treated as standard master nodes, and standard physical boundary conditions are applied directly without any modification.

Under the 2 : 1 balance condition (which is always maintained), the kinematics of these nodes is defined as:

- **Kinematic Motion:** A master node p has position $\boldsymbol{X}_p(t)$ and velocity $\boldsymbol{U}_p$. A hanging node h has position $\boldsymbol{X}_h(t)$ and velocity $\boldsymbol{U}_h$, constrained to remain at the midpoint of the segment connecting p_1 and p_2.
- **Half-Edge Notation:** For any node $k \in \mathcal{P}(c) \cup \mathcal{H}(c)$ of cell c, the subscripts $\overline{k}$ and $\underline{k}$ denote subcell half-edge segments in the counterclockwise and clockwise directions along $\partial\Omega_c$. For the collinear segment ph shown in Fig. 2, the unit outward normals coincide: $\boldsymbol{N}^c_{\underline{p}} = \boldsymbol{N}^c_{\overline{h}}$. The half-edge lengths adapt to the subdivided interface; for example, $L^c_{\underline{p}} = \frac{1}{2}L_{ph}$ and $L^c_{\overline{h}} = \frac{1}{2}L_{ph}$ for the parent cell. Subcell pressures for master nodes $(\Pi^c_{\overline{p}}, \Pi^c_{\underline{p}})$ and hanging nodes $(\pi^c_{\overline{h}}, \pi^c_{\underline{h}})$ are defined analogously to the conforming case.

For a cell c adjacent to a non-conforming interface, the set of nodal degrees of freedom is $\mathcal{P}(c) \cup \mathcal{H}(c)$, where $\mathcal{P}(c)$ contains the regular vertices and $\mathcal{H}(c)$ contains the hanging nodes belonging to cell c. In this way, cell c is regarded as a general polygon. For the coarse parent cell the hanging node h lies at the midpoint of an edge, whereas for each fine child cell it is one of the corners. In both cases h is a kinematically constrained node belonging to $\mathcal{H}(c)$, and each adjacent cell receives both the subcell hydrodynamic force $\boldsymbol{F}_{ch}$ and the constraint force λ_{ch} at h.

3.2. Decoupled kinematically constrained formulation

3.2.1. Unconstrained hanging-node functional

For a hanging node h, the subcell half-edge pressures are defined analogously to Eq. (4):

$$\begin{aligned} \pi^c_{\overline{h}} &= P_c + Z_c(\boldsymbol{U}_h - \boldsymbol{U}_c)\cdot \boldsymbol{N}^c_{\overline{h}}, \\ \pi^c_{\underline{h}} &= P_c + Z_c(\boldsymbol{U}_h - \boldsymbol{U}_c)\cdot \boldsymbol{N}^c_{\underline{h}}. \end{aligned} \tag{12}$$

The corresponding subcell and nodal matrices are:

$$\begin{gathered} \mathsf{M}^c_h = Z_c\left(L^c_{\overline{h}}\boldsymbol{N}^c_{\overline{h}} \otimes \boldsymbol{N}^c_{\overline{h}} + L^c_{\underline{h}}\boldsymbol{N}^c_{\underline{h}} \otimes \boldsymbol{N}^c_{\underline{h}}\right), \quad \mathsf{M}_h = \sum_{c\in\mathcal{C}(h)} \mathsf{M}^c_h, \\ L^c_h\boldsymbol{N}^c_h = L^c_{\overline{h}}\boldsymbol{N}^c_{\overline{h}} + L^c_{\underline{h}}\boldsymbol{N}^c_{\underline{h}}. \end{gathered} \tag{13}$$

If unconstrained, the hanging-node velocity would minimize the quadratic functional:

$$\mathcal{J}(\boldsymbol{U}_h) = \frac{1}{2}\boldsymbol{U}_h \cdot \left(\mathsf{M}_h\boldsymbol{U}_h\right) - \boldsymbol{U}_h \cdot \sum_{c\in\mathcal{C}(h)} \left(\mathsf{M}^c_h\boldsymbol{U}_c - P_c L^c_h \boldsymbol{N}^c_h\right), \tag{14}$$

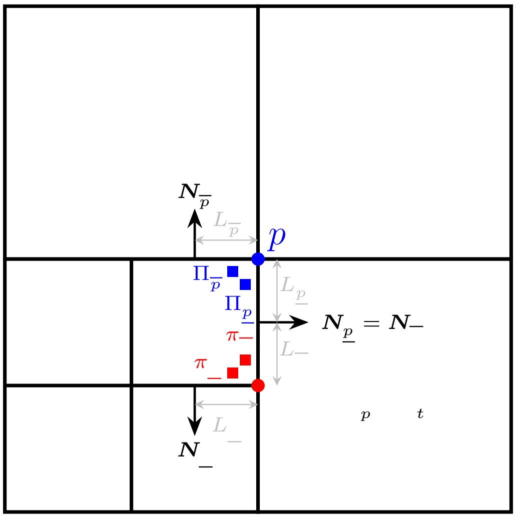


Figure 2: Schematic of a non-conforming interface between a coarse parent cell Ω_{parent} and two fine child cells. The hanging node h is constrained to remain collinear with master nodes p_1 and p_2. Half-edge notations for all three cells sharing the interface are indicated.

yielding the unconstrained velocity:

$$\boldsymbol{U}_h^{\text{free}} = \mathsf{M}_h^{-1} \sum_{c \in \mathcal{C}(h)} \left(\mathsf{M}_h^c \boldsymbol{U}_c - P_c L_h^c \boldsymbol{N}_h^c \right). \tag{15}$$

3.2.2. *Kinematic collinearity constraint*

To maintain geometric continuity, the hanging node must remain at the midpoint of its two adjacent master nodes p_1 and p_2. This imposes the holonomic geometric constraint:

$$\boldsymbol{C}(\boldsymbol{X}_h) \equiv \boldsymbol{X}_h - \frac{1}{2}\left(\boldsymbol{X}_{p_1} + \boldsymbol{X}_{p_2}\right) = \boldsymbol{0}, \tag{16}$$

which, upon differentiation with respect to time, yields the kinematic velocity constraint:

$$\boldsymbol{C}(\boldsymbol{U}_h) \equiv \boldsymbol{U}_h - \frac{1}{2}\left(\boldsymbol{U}_{p_1} + \boldsymbol{U}_{p_2}\right) = \boldsymbol{0}. \tag{17}$$

3.2.3. *Decoupled computation of the constraint force*

To enforce the constraint, we introduce a vector of Lagrange multipliers $\lambda_h \in \mathbb{R}^2$ and construct the Lagrangian function:

$$\mathcal{L}(\boldsymbol{U}_h, \lambda_h) = \mathcal{J}(\boldsymbol{U}_h) + \lambda_h \cdot \boldsymbol{C}(\boldsymbol{U}_h). \tag{18}$$

Substituting the explicit expressions yields:

$$\mathcal{L}(\boldsymbol{U}_h, \lambda_h) = \frac{1}{2}\boldsymbol{U}_h \cdot \left(\mathsf{M}_h \boldsymbol{U}_h\right) - \boldsymbol{U}_h \cdot \sum_{c \in \mathcal{C}(h)} \left(\mathsf{M}_h^c \boldsymbol{U}_c - P_c L_h^c \boldsymbol{N}_h^c \right) + \lambda_h \cdot \left[\boldsymbol{U}_h - \frac{1}{2}\left(\boldsymbol{U}_{p_1} + \boldsymbol{U}_{p_2}\right) \right]. \tag{19}$$

A key simplification arises because the constraint equation does not involve λ_h. With the master-node velocities already known from the baseline nodal solver (6), the hanging-node velocity is fixed by the kinematic condition alone. The Lagrange multiplier is then obtained directly from the force balance equation. In effect, this is a projection method: $\boldsymbol{U}_h$ is set to its kinematically constrained value, and λ_h is the residual force required to enforce collinearity.

The first-order necessary conditions for the stationary point of $\mathcal{L}$ are:

$$\frac{\partial \mathcal{L}}{\partial \lambda_h} \equiv \boldsymbol{U}_h - \frac{1}{2}\left(\boldsymbol{U}_{p_1} + \boldsymbol{U}_{p_2}\right) = \boldsymbol{0}, \tag{20a}$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{U}_h} \equiv \mathsf{M}_h \boldsymbol{U}_h - \sum_{c \in \mathcal{C}(h)} \left(\mathsf{M}_h^c \boldsymbol{U}_c - P_c L_h^c \boldsymbol{N}_h^c \right) + \lambda_h = \mathbf{0}. \tag{20b}$$

Equation (20a) gives the hanging-node velocity directly:

$$\boldsymbol{U}_h = \frac{1}{2}\left(\boldsymbol{U}_{p_1} + \boldsymbol{U}_{p_2}\right). \tag{21}$$

Substituting this into Eq. (20b) yields the constraint force:

$$\lambda_h = -\mathsf{M}_h \boldsymbol{U}_h + \sum_{c \in \mathcal{C}(h)} \left(\mathsf{M}_h^c \boldsymbol{U}_c - P_c L_h^c \boldsymbol{N}_h^c \right). \tag{22}$$

Here, λ_h is interpreted as the kinematic constraint force required to maintain the collinearity of the interface, thereby mitigating unphysical geometric incompatibilities such as mesh tearing, gaps, or overlap.

The *a posteriori* computation of λ_h has a clear geometric interpretation. Since the constraint manifold $\{\boldsymbol{U}_h = \frac{1}{2}(\boldsymbol{U}_{p_1} + \boldsymbol{U}_{p_2})\}$ is a single point, the constrained velocity $\boldsymbol{U}_h$ is fixed by the kinematic condition alone and does not depend on the unconstrained velocity $\boldsymbol{U}_h^{\text{free}}$ from Eq. (15).

$$\lambda_h = \mathsf{M}_h \left(\boldsymbol{U}_h^{\text{free}} - \boldsymbol{U}_h \right) \tag{23}$$

The Lagrange multiplier is then the generalized force that corrects the unconstrained velocity onto the constraint manifold. Because $\mathcal{J}$ is quadratic and the constraint is linear, this projection solves the local constrained minimization problem exactly for a single hanging node when the master-node velocities are held fixed. Globally, however, the scheme is a fractional-step strategy: master nodes are solved first and the hanging-node constraint is enforced *a posteriori*, so the decoupled update is not equivalent to solving the fully coupled global saddle-point system.

Remark on the cell-centered closure of constraint forces. In CCH, nodes carry no mass and act purely as kinematic entities for interface fluxes [11, 12]; unlike FEM or SGH nodes, they carry no independent momentum. Applying the constraint reactions directly to master nodes p_1, p_2 would couple the master-node solves back into a global saddle-point system. The constraint force λ_h must therefore act on cell centers, distributed as λ_{ch} via Eq. (33).

The updated cell-centered states re-enter the nodal solver at t^{n+1}, transmitting the constraint reaction to all adjacent nodes—including p_1 and p_2—through the usual cell→node→cell coupling. This fractional-step closure keeps the system decoupled and non-singular: each master-node matrix is positive-definite with $Z_c = \rho_c a_c > 0$, the hanging-node velocity is fixed by Eq. (20a), and λ_h then follows from Eq. (20b), without assembling a global saddle-point system.

3.3. Conservation properties on non-conforming meshes

The constraint force λ_h at hanging node h must be distributed to the adjacent cells. Let

$$\boldsymbol{F}_{ch} = \pi^c_{\overline{h}} L^c_{\overline{h}} \boldsymbol{N}^c_{\overline{h}} + \pi^c_{\underline{h}} L^c_{\underline{h}} \boldsymbol{N}^c_{\underline{h}} \tag{24}$$

denote the subcell hydrodynamic force at the hanging node. The semi-discrete equations for a cell adjacent to a non-conforming interface are:

$$m_c \frac{d}{dt}\left(\frac{1}{\rho_c}\right) - \sum_{p \in \mathcal{P}(c)} L_p^c \boldsymbol{N}_p^c \cdot \boldsymbol{U}_p - \sum_{h \in \mathcal{H}(c)} L_h^c \boldsymbol{N}_h^c \cdot \boldsymbol{U}_h = 0, \tag{25a}$$

$$m_c \frac{d\boldsymbol{U}_c}{dt} + \sum_{p \in \mathcal{P}(c)} \boldsymbol{F}_{cp} + \sum_{h \in \mathcal{H}(c)} \left(\boldsymbol{F}_{ch} + \lambda_{ch} \right) = \mathbf{0}, \tag{25b}$$

$$m_c \frac{dE_c}{dt} + \sum_{p \in \mathcal{P}(c)} \boldsymbol{F}_{cp} \cdot \boldsymbol{U}_p + \sum_{h \in \mathcal{H}(c)} \left(\boldsymbol{F}_{ch} + \lambda_{ch} \right) \cdot \boldsymbol{U}_h = 0. \tag{25c}$$

3.3.1. Mass conservation

No mass fluxes cross grid boundaries; m_c is constant for every cell. Global mass conservation holds trivially.

3.3.2. Geometric Conservation Law (GCL)

For a cell without hanging nodes, the rate of volume change is [11]:

$$\frac{dV_c}{dt} = \sum_{p\in\mathcal{P}(c)} L_p^c \boldsymbol{N}_p^c \cdot \boldsymbol{U}_p. \tag{26}$$

By treating each hanging node as an additional vertex, the cell is viewed as a general polygon, and its rate of volume change extends to:

$$\frac{dV_c}{dt} = \sum_{k\in\mathcal{P}(c)\cup\mathcal{H}(c)} L_k^c \boldsymbol{N}_k^c \cdot \boldsymbol{U}_k. \tag{27}$$

Since the cell volume is related to density by $V_c = m_c/\rho_c$, Eq. (25a) is equivalent to Eq. (27). In the CCH framework of Maire et al. [11], this equivalence is necessary and sufficient for GCL satisfaction. Therefore, the formulation satisfies the GCL at the semi-discrete level.

3.3.3. Momentum conservation

Summing Eq. (25b) over all cells and interchanging the order of summation yields:

$$\sum_c m_c \frac{d\boldsymbol{U}_c}{dt} + \sum_p \sum_{c\in\mathcal{C}(p)} \boldsymbol{F}_{cp} + \sum_h \sum_{c\in\mathcal{C}(h)} \left(\boldsymbol{F}_{ch} + \boldsymbol{\lambda}_{ch}\right) = \boldsymbol{0}. \tag{28}$$

A sufficient condition for global momentum conservation is that the nodal force sums vanish:

$$\sum_{c\in\mathcal{C}(p)} \boldsymbol{F}_{cp} = \boldsymbol{0}, \tag{29a}$$

$$\sum_{c\in\mathcal{C}(h)} \left(\boldsymbol{F}_{ch} + \boldsymbol{\lambda}_{ch}\right) = \boldsymbol{0}. \tag{29b}$$

For a regular node p, substituting the half-edge pressures and using Eq. (6) gives:

$$\sum_{c\in\mathcal{C}(p)} \boldsymbol{F}_{cp} = \mathsf{M}_p \boldsymbol{U}_p - \sum_{c\in\mathcal{C}(p)} \left(\mathsf{M}_p^c \boldsymbol{U}_c - P_c L_p^c \boldsymbol{N}_p^c\right) = \boldsymbol{0}. \tag{30}$$

For a hanging node h, using $\sum_c \boldsymbol{\lambda}_{ch} = \boldsymbol{\lambda}_h$ and Eq. (20b) yields:

$$\sum_{c\in\mathcal{C}(h)} \left(\boldsymbol{F}_{ch} + \boldsymbol{\lambda}_{ch}\right) = \mathsf{M}_h \boldsymbol{U}_h - \sum_{c\in\mathcal{C}(h)} \left(\mathsf{M}_h^c \boldsymbol{U}_c - P_c L_h^c \boldsymbol{N}_h^c\right) + \boldsymbol{\lambda}_h = \boldsymbol{0}. \tag{31}$$

Consequently, the rate of change of global momentum vanishes, and momentum is conserved.

3.3.4. Total energy conservation

Summing Eq. (25c) over all cells gives:

$$\sum_c m_c \frac{dE_c}{dt} + \sum_p \Big(\sum_{c\in\mathcal{C}(p)} \boldsymbol{F}_{cp} \Big) \cdot \boldsymbol{U}_p + \sum_h \Big(\sum_{c\in\mathcal{C}(h)} \left(\boldsymbol{F}_{ch} + \boldsymbol{\lambda}_{ch}\right) \Big) \cdot \boldsymbol{U}_h = 0. \tag{32}$$

Since the nodal velocities $\boldsymbol{U}_p$ and $\boldsymbol{U}_h$ are single-valued, they can be factored out of the cell-centered summations. Using the force balance conditions (29), the second and third terms in Eq. (32) vanish, yielding $\sum_c m_c \frac{dE_c}{dt} = 0$, which proves that global total energy is conserved.

3.3.5. Distribution of the constraint force

Introducing scalar partition coefficients α_{ch}:

$$\boldsymbol{\lambda}_{ch} = \alpha_{ch} \boldsymbol{\lambda}_h, \qquad \sum_{c\in\mathcal{C}(h)} \alpha_{ch} = 1. \tag{33}$$

Any partition summing to unity preserves global conservation. However, different choices of weights lead to substantial differences in thermodynamic robustness near strong discontinuities. To overcome the cold-cell bottleneck that arises with naive equal-weight or mass-weighted partitions, we define the partition coefficients based on the internal energy $m_c e_c = m_c E_c - \frac{1}{2} m_c |\boldsymbol{U}_c|^2$, which is a thermodynamic state variable invariant under Galilean transformations ($\boldsymbol{U} \to \boldsymbol{U} + \boldsymbol{V}_0$):

$$\alpha_{ch} = \frac{m_c e_c}{\sum_{k \in C(h)} m_k e_k}. \tag{34}$$

Let $W_h = \lambda_h \cdot \boldsymbol{U}_h$ denote the constraint power at node h. For an explicit time integration with step Δt, the constraint work extracted from cell c is $\Delta t\, \alpha_{ch} W_h$, and the internal energy evolves as

$$m_c e_c^{n+1} = m_c e_c^n - \Delta t\, \alpha_{ch} W_h + \cdots, \tag{35}$$

where the omitted terms collect the standard hydrodynamic fluxes common to all partitions. Analyzing the isolated constraint-work contribution reveals why different partitions lead to substantially different robustness. To avoid unphysical negative internal energy from constraint work extraction (when $W_h > 0$), one requires $e_c^{n+1} > 0$, yielding the per-cell condition $\alpha_{ch} < m_c e_c^n / (\Delta t\, W_h)$. Summing over $c \in C(h)$ gives the global necessary condition $\Delta t\, W_h < S_h^n$ with $S_h^n \equiv \sum_k m_k e_k^n$. This isolated constraint-work analysis establishes worst-case sufficient conditions under the isolated constraint-work model; it is a robustness rationale for the weighting choice, not a formal guarantee that the full update remains physically admissible.

For the internal-energy weighting (34), substituting $\alpha_{ch} = m_c e_c^n / S_h^n$ reduces all per-cell conditions to the single global bound $\Delta t\, W_h < S_h^n$. In contrast, equal-weight ($\alpha_{ch} = 1/|C(h)|$) and mass-weighted ($\alpha_{ch} = m_c / \sum_k m_k$) partitions impose the stricter per-cell restrictions $\Delta t\, W_h < |C(h)| \cdot \min_c (m_c e_c^n)$ and $\Delta t\, W_h < (\sum_k m_k) \cdot \min_c e_c^n$, respectively—each bottlenecked by the coldest adjacent cell. The internal-energy-weighted bound S_h^n is therefore the least restrictive of the three:

$$\max\left\{ |C(h)| \cdot \min_c (m_c e_c^n),\ \left(\sum\nolimits_k m_k\right) \cdot \min_c e_c^n \right\} \le S_h^n, \tag{36}$$

with strict inequality whenever the adjacent cells have heterogeneous internal energies—precisely the situation at a refinement interface crossed by a shock or material discontinuity. The internal-energy-weighted partition therefore assigns constraint work in proportion to each cell's capacity to supply it without becoming unphysical, which mitigates the cold-cell bottleneck and improves thermodynamic robustness near strong discontinuities. This motivates the robust performance observed in Sections 4 and 5, where it markedly reduces the occurrence of negative internal energies compared to equal-weight or mass-weighted alternatives. Since $\sum_k m_k e_k$ is strictly positive for all physically admissible thermodynamic states, the partition (34) is always well-defined; in the limit $e_c \to 0$, $\alpha_{ch} \to 0$ and the cell simply contributes no internal energy to the partition—consistent with the robustness rationale above. If any adjacent cell develops a negative internal energy, the partition loses its sign-definite interpretation and a fallback weighting (e.g., mass weighting) is used.

3.4. Time integration

The semi-discrete equations are advanced in time using an explicit first-order forward Euler scheme. The update sequence from time level n to $n+1$ is as follows:

1. **Geometry initialization:** Compute the subcell half-edge lengths and unit outward normals ($L_k^c, \boldsymbol{N}_k^c$) for all nodes $k \in \mathcal{P}(c) \cup \mathcal{H}(c)$ using the current node positions $\boldsymbol{X}^n$.
2. **Standard nodal solver:** For each regular node p, compute the velocity $\boldsymbol{U}_p^n$ via Eq. (6) using the cell-centered states ($\boldsymbol{U}_c^n, P_c^n$).
3. **Hanging nodal solver:** For each hanging node h, set the velocity to $\boldsymbol{U}_h^n = \frac{1}{2}(\boldsymbol{U}_{p_1}^n + \boldsymbol{U}_{p_2}^n)$ and compute the constraint force λ_h^n via Eq. (22).
4. **State update:** Distribute the constraint force as $\lambda_{ch}^n = \alpha_{ch} \lambda_h^n$ using Eq. (34), and update the cell velocity and specific total energy via Eqs. (25b)–(25c).
5. **Coordinate update:** Update the coordinates of all nodes: $\boldsymbol{X}_k^{n+1} = \boldsymbol{X}_k^n + \Delta t^n \boldsymbol{U}_k^n$.
6. **Thermodynamic update:** Calculate the new cell volume V_c^{n+1} from the updated coordinates $\boldsymbol{X}^{n+1}$, update the cell density $\rho_c^{n+1} = m_c / V_c^{n+1}$, and determine the new pressure P_c^{n+1} using the equation of state.

Table 1
Fixed-topology non-conforming mesh configurations. For each test case, the base resolution level (with equivalent conforming grid size), prescribed refined region, and refinement levels are specified.

Test Case	Base Level (Grid)	Refined Region	Refinement Level(s)
Sod	Level 4 (16×16)	$r \in [0.4, 0.8]$	Level 7 (128×128)
Sedov (A)	Level 5 (32×32)	$r < 0.25$ (L7), $0.25 \leq r < 0.5$ (L6)	L6, L7
Sedov (B)	Level 5 (32×32)	$x < 0.25$ (L7), $0.25 \leq x < 0.5$ (L6)	L6, L7
Taylor–Green	$L \in \{3, 4, 5, 6, 7\}$	$r \in [0.1, 0.4]$ centered at $(0.5, 0.5)$	$L + 1$

This time-marching procedure is consistent with the baseline cell-centered Lagrangian framework. The constraint force computed at step 3 influences the nodal solver at the next time step through the updated cell states. Because the semi-discrete conservation laws are integrated directly and the nodal force balances (29) hold exactly at every time step, global total energy is conserved to round-off in the fully discrete scheme; the machine-precision energy errors reported in Section 5.2 are therefore expected rather than coincidental.

3.4.1. Stability criteria

The time step Δt^n is limited by the most restrictive of the following criteria: (i) the standard Courant-Friedrichs-Lewy (CFL) condition with a CFL number $C_{\text{CFL}} = 0.2$; (ii) a constraint limiting the relative cell volume change to 10% per step; and (iii) a growth limit restricting the time step increase to $\Delta t^{n+1} \leq 1.1\Delta t^n$. Empirically, $C_{\text{CFL}} = 0.2$ has proven stable across all configurations tested in Section 4, including cases with multiple adjacent hanging nodes.

4. Numerical Validation on Fixed-Topology Non-Conforming Meshes

To validate the proposed hanging nodal solver using the ideal gas EOS, we conduct simulations on classical benchmark problems using the fixed-topology non-conforming meshes summarized in Table 1. Prescribing the grid topology *a priori* isolates the hanging-node treatment from dynamic remapping effects, enabling a systematic grid convergence study and direct comparison against uniform fine-mesh reference solutions. This setup aims to demonstrate that the solver: (i) conserves total energy; (ii) transmits waves across non-conforming interfaces with minimal spurious reflections or symmetry breaking; (iii) achieves accuracy comparable to uniform fine meshes; and (iv) exhibits the expected asymptotic convergence rate.

4.1. Sod

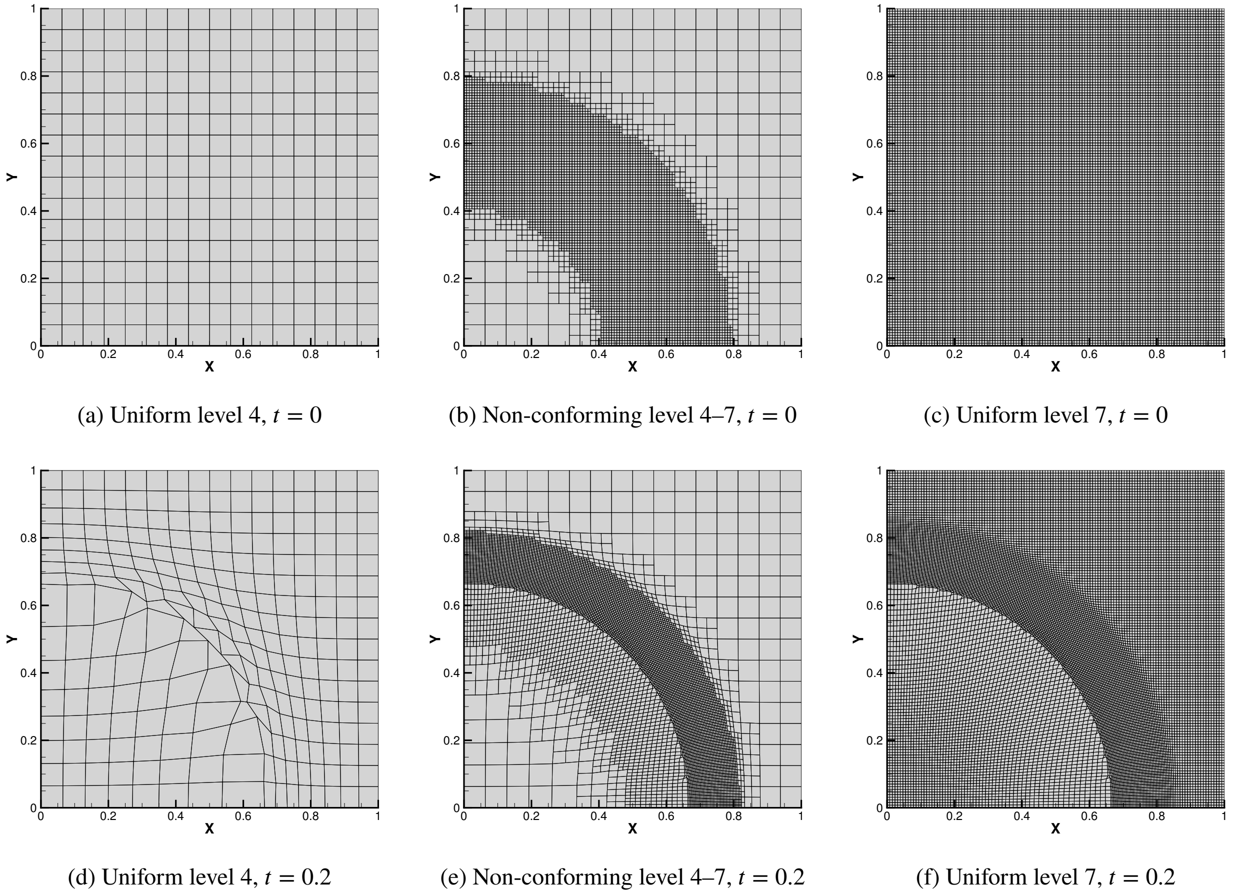
We consider a quarter-domain cylindrical Sod problem with the origin located at the lower-left corner and reflective boundaries enforcing symmetry. The initial circular discontinuity is at radius $r = \sqrt{x^2 + y^2} = 0.5$. The initial state is

$$(\rho, P, u, v) = \begin{cases} (1.0,\ 1.0,\ 0,\ 0), & r < 0.5, \\ (0.125,\ 0.1,\ 0,\ 0), & r \geq 0.5, \end{cases} \tag{37}$$

with the ideal gas equation of state ($\gamma = \frac{5}{3}$). The simulation is advanced to $t = 0.2$. Because the cylindrical Sod problem lacks an exact analytical solution, a high-resolution numerical solution obtained using a 1000×1000 uniform Eulerian mesh serves as the reference solution. Owing to its far finer resolution, the numerical error of this reference is negligible compared with that of the level 4–7 meshes under study, so it provides a reliable benchmark despite being computed with a different (Eulerian) discretization.

We employ three mesh configurations to assess the performance of the hanging nodal solver:

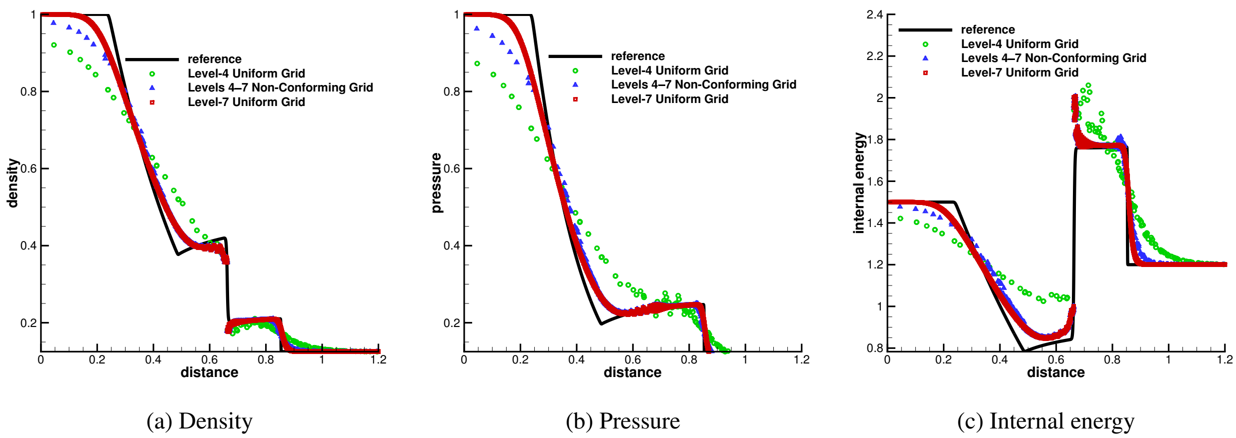
1. **Uniform level 4 mesh** (16×16 cells)—coarse baseline;
2. **Uniform level 7 mesh** (128×128 cells)—fine resolution;
3. **Non-conforming level 4–7 mesh**—base level 4 with three additional refinement levels (level 7) applied in the annular band $0.4 \leq r \leq 0.8$, which encompasses the initial discontinuity and the evolving shock, contact, and rarefaction waves.



(a) Uniform level 4, $t = 0$ (b) Non-conforming level 4–7, $t = 0$ (c) Uniform level 7, $t = 0$

(d) Uniform level 4, $t = 0.2$ (e) Non-conforming level 4–7, $t = 0.2$ (f) Uniform level 7, $t = 0.2$

Figure 3: Initial (top row) and final at $t = 0.2$ (bottom row) meshes for the cylindrical Sod problem with three configurations. The non-conforming mesh (middle column) refines the annular region $0.4 \leq r \leq 0.8$ from level 4 to level 7, placing both the initial discontinuity and the principal wave structures within the refined zone.



(a) Density (b) Pressure (c) Internal energy

Figure 4: Radial profiles of density, pressure, and specific internal energy at $t = 0.2$ for the cylindrical Sod problem. Results from the uniform level 4 mesh, non-conforming level 4–7 mesh, and uniform level 7 mesh are compared against the Eulerian reference solution (1000×1000 uniform mesh).

Figure 3 shows the initial and final meshes. The non-conforming mesh maintains geometric continuity at both refinement interfaces ($r = 0.4$ and $r = 0.8$) throughout the deformation, and the mesh preserves cylindrical symmetry along all radial directions, confirming that the kinematic constraint introduces no asymmetric perturbations.

Figure 4 compares radial profiles against the high-resolution Eulerian reference solution. All three configurations capture the rarefaction, contact discontinuity, and shock front with correct positions and amplitudes. The uniform level 4 mesh exhibits the largest numerical diffusion; the uniform level 7 mesh provides sharper resolution. The non-conforming level 4–7 mesh yields accuracy intermediate between the two uniform meshes: **within the refined band** ($0.4 \le r \le 0.8$), the solution quality approaches that of the uniform level 7 mesh, while **outside the refined band** it retains level 4 resolution. Visually, the wave profiles transition smoothly across the refinement interfaces without significant spurious oscillations, confirming that the hanging nodal solver transmits waves correctly across the level boundaries.

4.2. Sedov blast wave

The Sedov problem [9] models a point-explosion blast wave in a quarter-circle $[0, 1.2] \times [0, 1.2]$ with reflective boundaries. The initial state is $\rho_0 = 1.0$, $P_0 = 10^{-6}$, with total internal energy 0.244816 deposited near the origin. The exact solution at $t = 1.0$ has peak density $\rho = 6$ at radius $r = 1.0$.

We employ three mesh configurations to evaluate the performance of the hanging nodal solver:

1. **Uniform level 5 mesh** (32×32 cells)—uniform baseline reference;
2. **Refined level 5–7 mesh A**—base level 5 with a smoothly graded transition: refined to level 7 in the central energy-deposition zone ($r < 0.25$), level 6 in the intermediate annular region ($0.25 \le r < 0.5$), and level 5 elsewhere;
3. **Refined level 5–7 mesh B**—base level 5 with a smoothly graded transition along the x-axis: refined to level 7 in the region $x < 0.25$, level 6 in the intermediate region $0.25 \le x < 0.5$, and level 5 elsewhere.

Figure 5 shows the initial and final meshes. Both non-conforming configurations produce shock wavefronts that maintain excellent circular symmetry, matching the uniform level 5 baseline. Hanging nodes remain collinear with their master nodes throughout the entire simulation, confirming that the kinematically constrained formulation enforces geometric continuity.

Figure 6 compares the radial density profiles against the exact solution. The exact peak density is $\rho = 6.0$ at $r = 1.0$. The uniform level 5 mesh captures the primary shock features correctly, attaining a peak density of approximately 5.2 (slightly below the exact peak due to numerical diffusion). The two non-conforming configurations yield results in close agreement with this uniform baseline, with mesh A reaching a similar peak density of approximately 5.2. Because the level 5–7 mesh B maintains a level 6 refinement at the shock front at the final simulation time, it achieves closer agreement with the exact solution, reaching a peak density of approximately 6.0.

4.3. Taylor–Green vortex

This case [8] describes a vortex-dominated flow. In Lagrangian hydrodynamics, the strong rotation causes severe grid distortion, posing a challenge for mesh preservation and solver stability.

The computational domain is defined as a unit square $\Omega = [0, 1] \times [0, 1]$ enclosed by wall boundaries. The fluid is modeled as an ideal gas with adiabatic index $\gamma = 1.4$. The initial distribution of the flow field—including velocity $\boldsymbol{U}_0 = (u_0, v_0)^T$, density ρ_0, and pressure P_0—is given by the following analytical expressions:

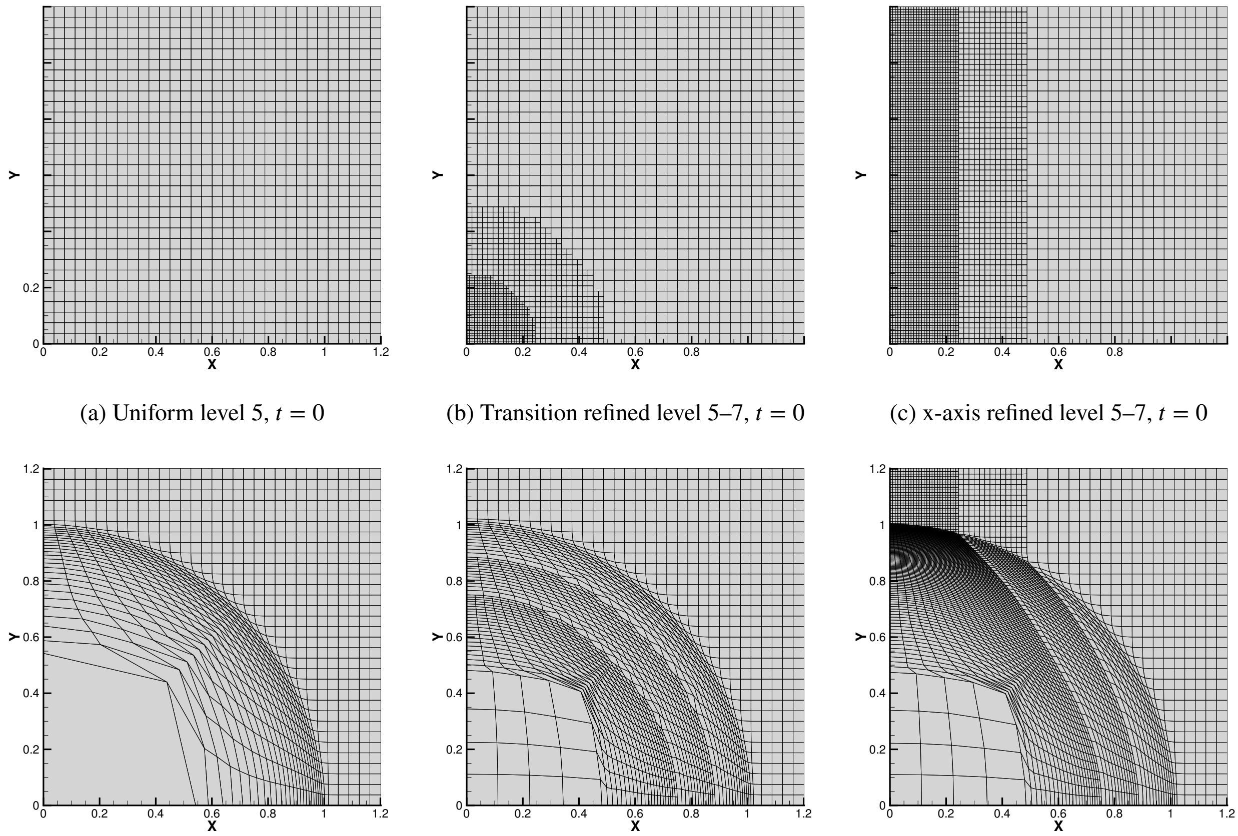
$$u_0(x, y) = \sin(\pi x)\cos(\pi y), \quad v_0(x, y) = -\cos(\pi x)\sin(\pi y),$$
$$\rho_0(x, y) = 1.0, \quad P_0(x, y) = \frac{1}{4}\rho_0\left[\cos(2\pi x) + \cos(2\pi y)\right] + 1.0.$$

To sustain a steady flow, a source term is added to the internal-energy update for this test case alone:

$$\overline{s}_e = \frac{5\pi}{8}\left[\cos(\pi x)\cos(3\pi y) - \cos(3\pi x)\cos(\pi y)\right].$$

Because this external forcing is not conservative, the Taylor–Green case does not test global energy conservation.

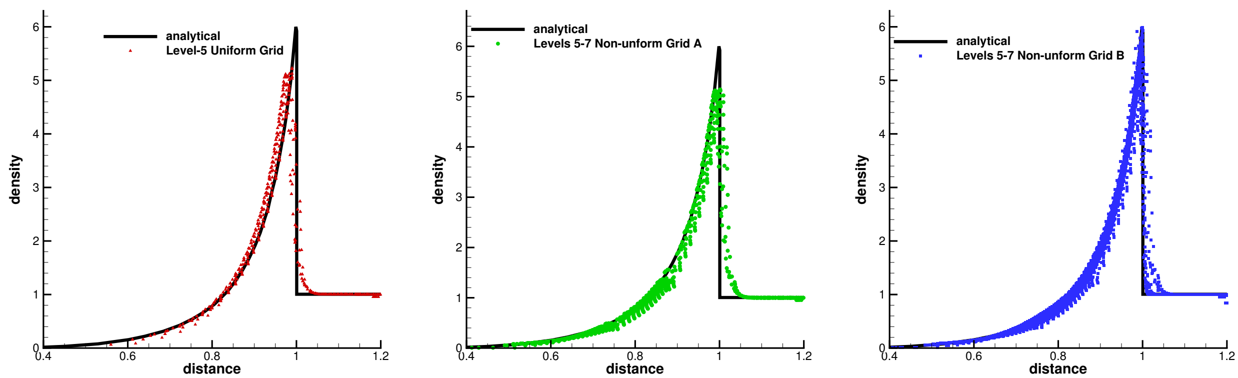
To evaluate the performance of the proposed method, both uniform meshes and non-conforming adaptive meshes are considered. We denote by h_0 the reference grid spacing at the base resolution level (Level 3). For the non-conforming meshes, a localized refinement zone is prescribed; specifically, cells within the base mesh of level L that



(a) Uniform level 5, $t = 0$ (b) Transition refined level 5–7, $t = 0$ (c) x-axis refined level 5–7, $t = 0$

(d) Uniform level 5, $t = 1.0$ (e) Transition refined level 5–7, $t = 1.0$ (f) x-axis refined level 5–7, $t = 1.0$

Figure 5: Initial (top row) and final at $t = 1.0$ (bottom row) meshes for the Sedov blast wave problem under three configurations. The non-conforming meshes (middle and right columns) refine the central energy-deposition zone and the blast wave propagation path from level 5 to level 7.



(a) Uniform level 5 (b) Transition refined vs. uniform (c) x-axis refined vs. uniform

Figure 6: Radial density profiles at $t = 1.0$ for the Sedov blast wave problem. Results from the three configurations are compared against the exact analytical solution.

Table 2
Grid convergence study for the Taylor–Green vortex problem at $t = 0.6$. The global L^2 errors for velocity and the corresponding convergence rates are compared for uniform and non-conforming meshes. The effective mesh size $h_{\text{eff}} = \sqrt{|\Omega|/N}$ is a global average; within the refined annular zone the local resolution is finer than h_{eff} suggests.

Uniform Meshes				Non-conforming Meshes			
Mesh	h_{eff}	$\|e_u\|_{L^2}$	Order	Mesh	h_{eff}	$\|e_u\|_{L^2}$	Order
Level 4	$h_0/2$	3.01×10^{-1}	—	Levels 4–5	3.88×10^{-2}	2.49×10^{-1}	—
Level 5	$h_0/4$	1.77×10^{-1}	0.77	Levels 5–6	2.02×10^{-2}	1.51×10^{-1}	0.77
Level 6	$h_0/8$	9.67×10^{-2}	0.87	Levels 6–7	1.01×10^{-2}	8.36×10^{-2}	0.85
Level 7	$h_0/16$	5.11×10^{-2}	0.92	Levels 7–8	5.03×10^{-3}	4.45×10^{-2}	0.91
Level 8	$h_0/32$	2.59×10^{-2}	0.98	Levels 8–9	2.51×10^{-3}	2.28×10^{-2}	0.96

fall into the annular region centered at $(0.5, 0.5)$ with radius satisfying $0.1 \leq r \leq 0.4$ are further refined to level $L + 1$. To facilitate a consistent comparison across different grid configurations, the effective global mesh size is defined as $h_{\text{eff}} = \sqrt{|\Omega|/N}$, where $|\Omega|$ is the domain area and N represents the total number of cells. The measure h_{eff} is a global average; for the non-conforming meshes, the local resolution is finer within the refined annular region ($0.1 \leq r \leq 0.4$) and coarser outside it.

We conduct a grid convergence study to evaluate the accuracy of the proposed hanging nodal solver under vortex-dominated flows. We simulate the Taylor–Green vortex problem up to $t = 0.6$. Table 2 reports the global L^2 errors of velocity and the corresponding convergence orders for both uniform and non-conforming meshes.

For the uniform meshes (Levels 4 to 8), the observed convergence order ranges from 0.77 to 0.98 as the mesh is refined. For the non-conforming meshes (Levels 4–5 to Levels 8–9), the observed convergence orders range from 0.77 to 0.96. The error magnitudes and convergence rates of the non-conforming meshes track those of the uniform meshes closely across all resolution levels, with differences of at most a few percent. This demonstrates that the hanging nodal solver introduces negligible additional numerical error at the non-conforming interfaces and that the kinematically constrained treatment does not degrade the inherent accuracy of the baseline scheme. We note that the non-conforming convergence orders are effective rates measured against the global average h_{eff}; because the error is dominated by the coarser regions outside the refined annular zone, they should not be interpreted as pointwise or local convergence rates.

5. Extension to Adaptive Mesh Refinement

The preceding sections validated the proposed hanging nodal solver on fixed-topology non-conforming meshes, isolating the interface physics from remapping errors. In this section, we demonstrate the integration of the proposed solver into an AMR framework, showing that the conservation properties and solver robustness are preserved under mesh refinement and coarsening.

5.1. Solver Coupling

To couple the Lagrangian solver with an adaptively refined grid, we employ the `p4est` library [5] to manage the logical quadtree topology and parallel communication. The cell-centered and nodal physical variables are stored as user data (payload) on the leaf quadrants.

The numerical operations of the Lagrangian scheme are mapped onto the native iterators of `p4est` as follows:

- **Corner Iterator**: Computes the velocities of all master nodes via Eq. (6).
- **Face Iterator**: Enforces the kinematic constraint on hanging nodes using Eq. (21) and computes the corresponding constraint forces λ_h via Eq. (22).
- **Volume Iterator**: Integrates the semi-discrete conservation laws (25) to update cell-centered physical states.

Mesh adaptation is triggered dynamically using gradient-based indicators of density or pressure, with cell refinement and coarsening adhering to standard quadtree $2:1$ balancing rules. During dynamic grid adaptation, physical states must be transferred upon topology changes. For mesh refinement, the child cells directly inherit the thermodynamic and kinematic states of their parent, and their new nodal coordinates are generated preserving the exact parent geometry, which is conservative at the semi-discrete level. For mesh coarsening, an intersection-based

Table 3
Mesh adaptation parameters for different benchmark cases, including base/maximum grid levels, adaptation criterion, and thresholds for refinement (ϵ_{ref}) and coarsening (ϵ_{coarse}).

Test Case	Base Level	Max Level	Criterion	ϵ_{ref}	ϵ_{coarse}
Sod	4	8	Density Gradient	1.0	0.8
Sedov	5	7	Density Gradient	8.0	6.0
2D Riemann	4	6	Density Gradient	8.0	7.6
Triple point	70×30	+2 levels	Density Gradient	8.0	7.6

conservative remap is employed to aggregate the conserved variables (mass, momentum, and total energy) from the fine child cells into a coarser parent cell, ensuring that they remain algebraically conserved up to machine precision:

$$m_P = \sum_i m_i, \quad (m\boldsymbol{U})_P = \sum_i m_i \boldsymbol{U}_i, \quad (mE)_P = \sum_i m_i E_i. \tag{38}$$

The parent cell's primitive variables are then derived as $\boldsymbol{U}_P = (m\boldsymbol{U})_P/m_P$ and $E_P = (mE)_P/m_P$, with the specific internal energy determined accordingly. Note that this first-order remap is conservative in mass, momentum, and total energy, but not in internal energy: the kinetic energy of sub-cell velocity fluctuations is partially converted into internal energy.

5.2. Numerical Validation

To verify the performance of the proposed solver coupled with AMR, we simulate four benchmark problems: the Sod shock tube, the Sedov blast wave, the two-dimensional Riemann problem, and the triple point problem. The grid levels and adaptation parameters for each case are summarized in Table 3.

The objective of these tests is to demonstrate that global physical conservation—specifically total energy conservation—is preserved under mesh refinement and coarsening. A first-order conservative remapping is applied during all quadtree-splitting and merging operations. While this remapping introduces local numerical diffusion, it is designed to preserve global conservation of mass, momentum, and total energy at the semi-discrete level. The AMR framework employed in this section serves primarily as a stress test for the non-conforming interface formulation. For all subsequent AMR test cases, the grid adaptation (refinement and coarsening) is deliberately triggered every 4 time steps. This strategy induces frequent topological changes, allowing us to test the robustness and the semi-discrete conservative properties of the scheme under severe conditions. Evaluating CPU efficiency or optimizing refinement criteria is deferred to future work.

5.2.1. Sod shock tube

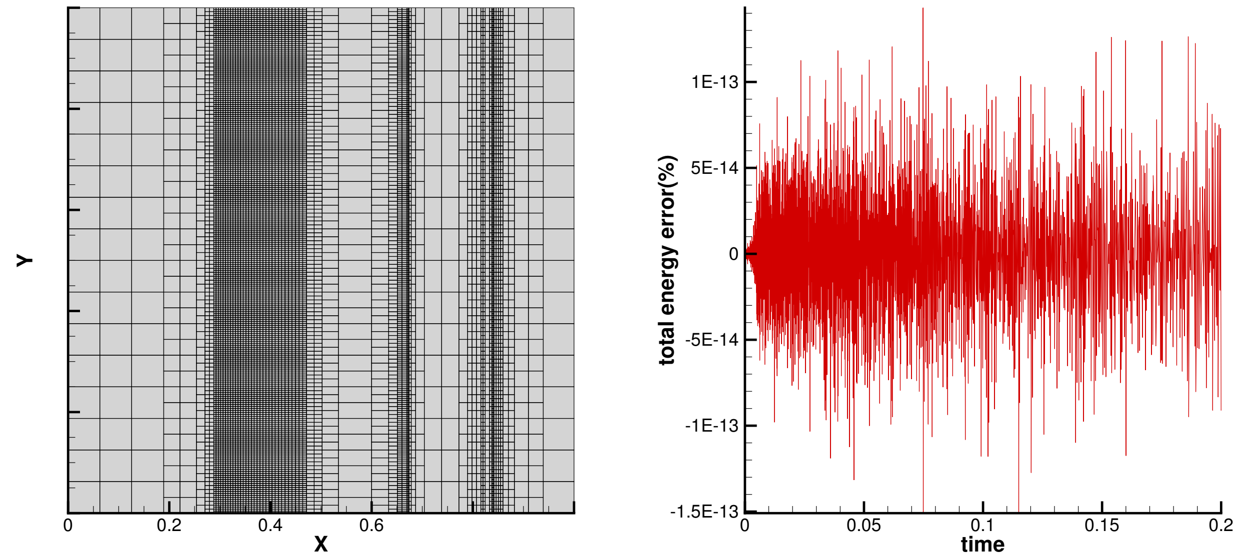
We simulate the Sod shock tube problem to verify the accuracy and symmetry preservation of the solver under mesh adaptation. The domain is a unit square $\Omega = [0, 1] \times [0, 1]$ with reflective wall boundaries. The working fluid is modeled as an ideal gas with $\gamma = 1.4$. The initial discontinuity is located at $x = 0.5$, with the initial states defined as:

$$(\rho, P, u, v) = \begin{cases} (1.0, 1.0, 0, 0), & \text{for } x < 0.5, \\ (0.125, 0.1, 0, 0), & \text{for } x \geq 0.5. \end{cases} \tag{39}$$

The simulation is advanced to a final time of $t = 0.2$. The AMR scheme clusters refined cells (up to level 8) around the shock front and the contact discontinuity, while maintaining a coarser grid in uniform flow regions (Fig. 7(a)). The computed density, pressure, and internal energy centerline profiles show close agreement with the exact analytical solution (Fig. 8). The dynamically adapted grid and the resulting physical fields maintain the one-dimensional symmetry along the flow direction throughout the calculation, confirming that the interface treatment introduces no multi-dimensional numerical artifacts. Despite the frequent topological variations induced by the grid adaptation, the relative total energy error stays at machine precision ($\sim 10^{-14}$). This indicates that the hanging nodal solver and the conservative remapping behave consistently and preserve global conservation during frequent grid modifications (Fig. 7(b)).

5.2.2. Sedov blast wave

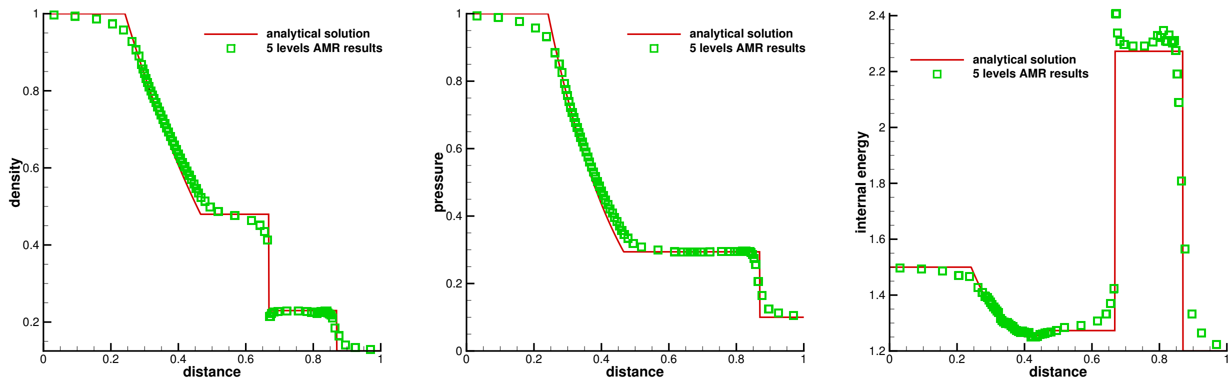
We simulate the Sedov point explosion to test the solver under strong, expanding shocks. The computational domain is a quarter-circle in $[0, 1.2]\times[0, 1.2]$ with reflective boundaries. The domain is initially filled with an ideal gas ($\gamma = 1.4$)



(a) Mesh distribution

(b) Total energy relative error

Figure 7: Numerical results for the Sod shock tube problem at $t = 0.2$: (a) adapted quadtree mesh distribution, where refined cells cluster around the shock front and the contact discontinuity; (b) time history of the total energy relative error, showing that total energy is well conserved.



(a) Density

(b) Pressure

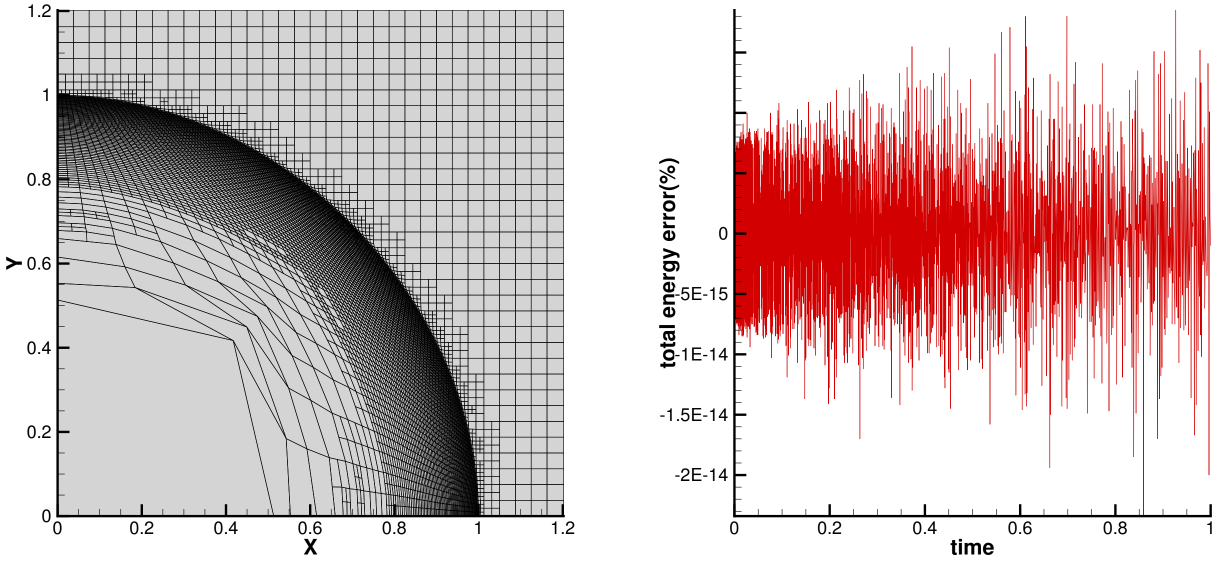
(c) Internal energy

Figure 8: Comparison of the numerical and exact profiles along the centerline for the Sod problem at $t = 0.2$: (a) density, (b) pressure, and (c) internal energy.

at rest, with a uniform density $\rho_0 = 1.0$ and background pressure $P_0 = 10^{-6}$. A total internal energy of 0.244816 is deposited in the cells adjacent to the origin. The simulation is advanced using an AMR configuration with levels 5–7 up to $t = 1.0$.

The grid adapts to follow the expanding shock wave (Fig. 9(a)). The computed density profile is compared with the exact analytical solution in Fig. 10. The shock peak is resolved, and despite the frequent topological variations, the relative total energy error stays at machine precision ($\sim 10^{-14}$), demonstrating consistent global conservation (Fig. 9(b)).

To numerically validate the practical robustness of the internal-energy-weighted partition strategy discussed in Section 3.3, we perform Sedov blast wave simulations using different partition strategies. While the equal-weight and mass-weighted approaches quickly encountered negative internal energies near the refinement interfaces, the proposed internal-energy-weighted strategy demonstrated significantly better robustness, with negative internal energies arising only very rarely, and in most test cases not at all.



(a) Adapted mesh

(b) Relative energy error

Figure 9: Sedov blast wave results under a level 5–7 AMR mesh: (a) adapted mesh, and (b) time history of the total energy relative error, confirming energy conservation near machine precision.

5.2.3. Two-dimensional Riemann problem

We simulate the two-dimensional Riemann problem to evaluate the solver's ability to resolve complex shock interactions and vortex structures under mesh adaptation. The domain is a unit square $\Omega = [0, 1] \times [0, 1]$ with reflective boundaries on all sides. The fluid is modeled as an ideal gas with $\gamma = 1.4$. The initial discontinuity is placed at $x = 0.5$ and $y = 0.5$, partitioning the domain into four quadrants initialized as:

$$
(\rho, P, u, v) = \begin{cases} (1.1,\ 1.1,\ 0.8939,\ 0.8939), & \text{for } x < 0.5,\ y < 0.5, \\ (0.5065,\ 0.35,\ 0.0,\ 0.8939), & \text{for } x > 0.5,\ y < 0.5, \\ (0.5065,\ 0.35,\ 0.8939,\ 0.0), & \text{for } x < 0.5,\ y > 0.5, \\ (1.1,\ 1.1,\ 0.0,\ 0.0), & \text{for } x > 0.5,\ y > 0.5. \end{cases} \tag{40}
$$

The simulation is run until the final time $t = 0.2$. Three meshes are compared: a level 4 coarse mesh, a level 6 fine mesh, and a level 4–6 AMR mesh.

Figure 11 presents the computed density fields. The level 4 mesh suffers from severe numerical diffusion, smearing the shock fronts and central vortices (Fig. 11(a)). In contrast, the level 4–6 AMR scheme (Fig. 11(b)) captures the

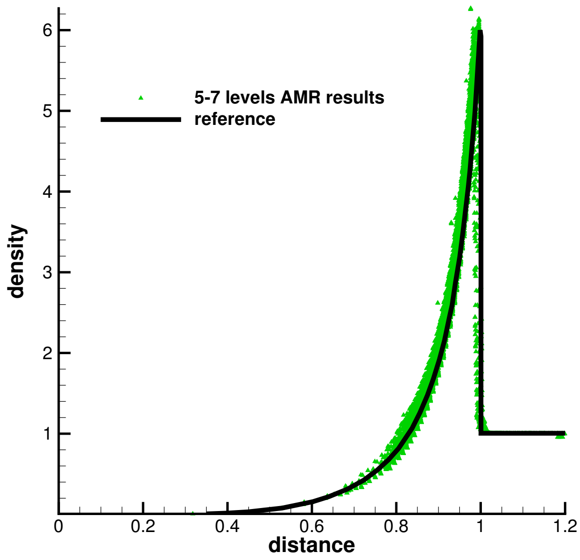
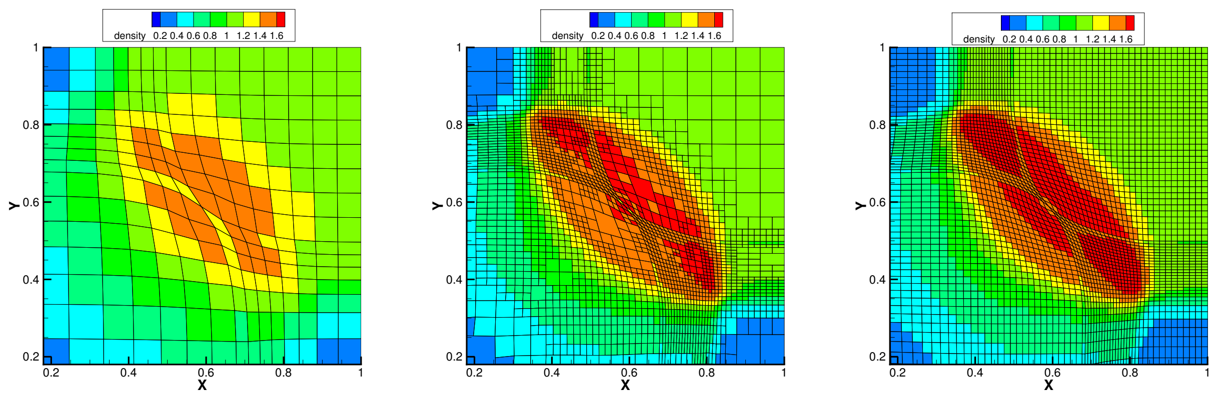


Figure 10: Comparison of the numerical and exact density profiles along the diagonal line for the Sedov problem at $t = 1.0$ under levels 5–7 AMR.

detailed shock interactions and vortex roll-ups with a fidelity comparable to that of the uniformly fine 6-level mesh (Fig. 11(c)), using significantly fewer cells. Diagonal symmetry along the principal line $x = y$ is well-preserved, confirming the isotropy of the nodal solver. Furthermore, total energy conservation is well maintained, with relative errors remaining below 10^{-14} (Fig. 12).



(a) Uniform level 4 mesh (b) AMR levels 4–6 (c) Uniform level 6 mesh

Figure 11: Density fields and grid distributions for the 2D Riemann problem at $t = 0.2$: (a) uniform level 4 mesh, (b) AMR (levels 4–6), and (c) uniform level 6 mesh.

5.2.4. Triple point problem

We simulate the triple point problem to verify the geometric robustness and conservation properties of the solver under extreme shear and material interface deformations. The domain is $[0, 7] \times [0, 3]$ with reflective boundaries on all sides. The working fluid is modeled as an ideal gas with $\gamma = 1.4$. The three material regions occupy the following subdomains at $t = 0$:

$$(\rho, P, u, v) = \begin{cases} (1.0,\ 1.0,\ 0.0,\ 0.0), & \text{for } x \le 1.0, \\ (1.0,\ 0.1,\ 0.0,\ 0.0), & \text{for } x > 1.0 \text{ and } y \le 1.5, \\ (0.125,\ 0.1,\ 0.0,\ 0.0), & \text{for } x > 1.0 \text{ and } y > 1.5. \end{cases} \tag{41}$$

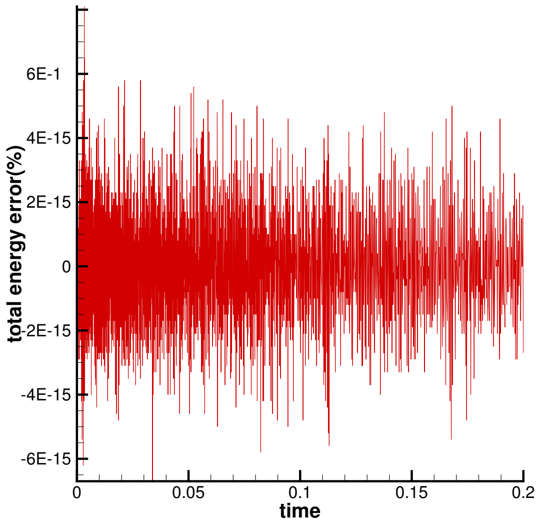


Figure 12: Time history of the total energy relative error for the 2D Riemann problem under AMR.

The initial grid is a 70×30 Cartesian mesh, with two additional refinement levels allowed during adaptation (+2 levels). The simulation is advanced to $t = 2.0$.

Figure 13 compares the base mesh and the dynamically adapted AMR mesh at $t = 2.0$. The AMR scheme effectively concentrates refined cells along the material interfaces and the vortex structures of the Kelvin–Helmholtz instabilities. A local enlargement of the adapted mesh (Fig. 14) confirms that the grid remains well-conditioned and free of tangling or negative volume collapse under severe shear. The resolved density field matches the reference results qualitatively (Fig. 15), and the total energy relative error remains bounded near machine precision (Fig. 16).

6. Conclusion and Future Work

In this paper, we have developed a cell-centered Lagrangian-AMR scheme based on a hanging nodal solver for non-conforming quadrilateral meshes. By exploiting the mathematical structure of the classical CCH nodal solver as a minimizer of a quadratic functional, we have proposed a decoupled kinematically constrained projection method. This method enforces the midpoint collinearity constraint at non-conforming interfaces by embedding it into the solver's variational framework. The key features and contributions of this work are summarized as follows:

1. **Decoupled Kinematically Constrained:** The kinematic constraints are decoupled from the active master node solvers. The velocities of master nodes are computed using the baseline solver, and the hanging node's velocity is kinematically constrained. The constraint force is computed *a posteriori* from the nodal momentum balance, avoiding the need to solve coupled systems.
2. **Conservative Force Redistribution:** To close the system, the constraint force is distributed back to adjacent cells using an internal-energy-weighted partition. This redistribution ensures the global conservation of mass, momentum, and total energy, while simultaneously satisfying the Geometric Conservation Law (GCL).
3. **Numerical Validation:**
    - On fixed-topology non-conforming meshes, tests on Sod shock tube and Sedov blast wave problems verify that the solver preserves cylindrical symmetry and transmits waves correctly across refinement interfaces. A Taylor–Green vortex convergence study demonstrates that non-conforming meshes achieve error levels and convergence rates comparable to those of uniform meshes, indicating that the hanging-node treatment does not degrade the accuracy of the baseline scheme.
    - Under Adaptive Mesh Refinement (AMR) coupled with the `p4est` library, benchmarks such as the two-dimensional Riemann problem and the triple point problem demonstrate that the solver is robust under severe grid deformation and shear flows. The scheme maintains global conservation of total energy throughout all mesh adaptation and remapping cycles.

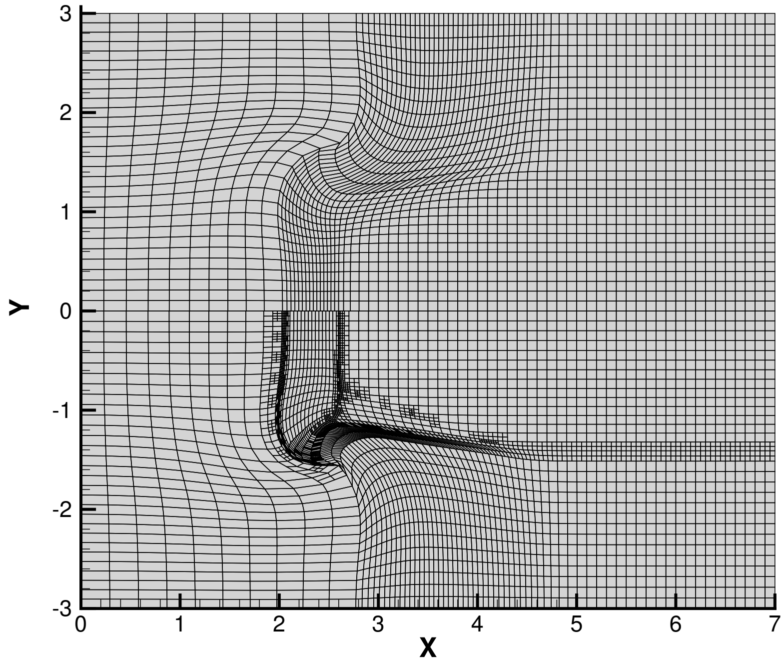


Figure 13: Comparison of the base mesh (top) and the dynamically adapted AMR mesh (bottom) for the triple point problem at $t = 2.0$.

- Maintaining smooth gradient-based mesh transitions along shock propagation paths appears to suppress numerical oscillations in strong shock problems; we defer a quantitative study of this effect to future work.

Future extensions include raising spatial and temporal accuracy to second order to reduce numerical diffusion, extending the formulation to the ALE framework for large-deformation problems, and establishing unified benchmark metrics for systematic cross-code comparison with alternative non-conforming Lagrangian schemes.

Acknowledgements

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References

[1] Anderson, R., Elliott, N., Pember, R., 2004. An arbitrary Lagrangian–Eulerian method with adaptive mesh refinement for the solution of the Euler equations. Journal of Computational Physics 199, 598–617.

[2] Belytschko, T., Liu, W.K., Moran, B., Elkhodary, K., 2014. Nonlinear Finite Elements for Continua and Structures. 2nd ed., John Wiley & Sons.

[3] Berger, M.J., Colella, P., 1989. Local adaptive mesh refinement for shock hydrodynamics. Journal of Computational Physics 82, 64–84.

[4] Bernardi, C., Maday, Y., Patera, A.T., 1994. A mortar element method for third order elliptic equations. Applicable Analysis 53, 1–24.

[5] Burstedde, C., Wilcox, L.C., Ghattas, O., 2011. p4est: Scalable algorithms for parallel adaptive mesh refinement on forests of octrees. SIAM Journal on Scientific Computing 33, 1103–1133. doi:10.1137/100791634.

[6] Caramana, E., Burton, D., Shashkov, M.J., Whalen, P., 1998. The construction of compatible hydrodynamics algorithms utilizing conservation of total energy. Journal of Computational Physics 146, 227–262.

[7] Colaïtis, A., Guisset, S., Breil, J., 2026. A cell-centered AMR-ALE framework for 3D multi-material hydrodynamics. Part I: Lagrangian and indirect Euler AMR algorithms. Journal of Computational Physics , 114701.

[8] Dobrev, V.A., Kolev, T.V., Rieben, R.N., 2012. High-order curvilinear finite element methods for Lagrangian hydrodynamics. SIAM Journal on Scientific Computing 34, B606–B641.

[9] Kamm, J.R., Timmes, F.X., 2007. On efficient generation of numerically robust Sedov solutions. Technical Report. Technical Report LA-UR-07-2849, Los Alamos National Laboratory.

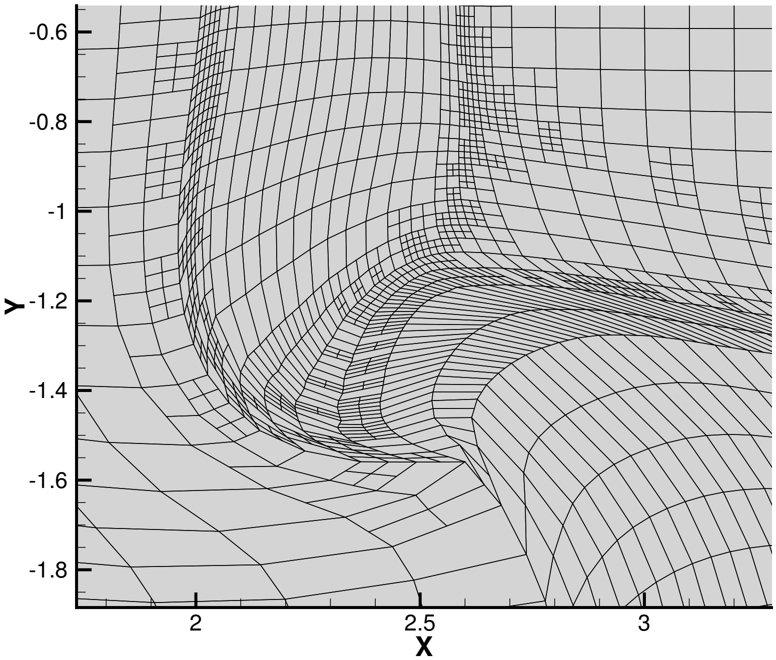

Figure 14: Local enlargement of the adapted quadtree mesh for the triple point problem at $t = 2.0$, showing grid quality and collinearity near sheared interfaces.

[10] Koniges, A., Anderson, R., Wang, P., Gunney, B., Becker, R., Eder, D., Mac McGowan, B., Schneider, M., 2006. Modeling NIF experimental designs with adaptive mesh refinement and Lagrangian hydrodynamics, in: Journal de Physique IV (Proceedings), EDP sciences. pp. 587–593.

[11] Maire, P.H., 2009. A high-order cell-centered Lagrangian scheme for two-dimensional compressible fluid flows on unstructured meshes. Journal of Computational Physics 228, 2391–2425.

[12] Maire, P.H., Abgrall, R., Breil, J., Ovadia, J., 2007. A cell-centered lagrangian scheme for two-dimensional compressible flow problems. SIAM Journal on Scientific Computing 29, 1781–1824.

[13] Morrell, J., Sweby, P., Barlow, A., 2007. A cell by cell anisotropic adaptive mesh ALE scheme for the numerical solution of the Euler equations. Journal of Computational Physics 226, 1152–1180.

[14] Wang, P., 2010. Modeling material responses by arbitrary Lagrangian Eulerian formulation and adaptive mesh refinement method. Journal of Computational Physics 229, 1573–1599.

[15] Wilkins, M.L., 1964. Calculation of elastic-plastic flow. Methods in Computational Physics: Advances in Research and Applications 3, 211–263.

[16] Wriggers, P., 2006. Computational Contact Mechanics. Springer Science & Business Media.

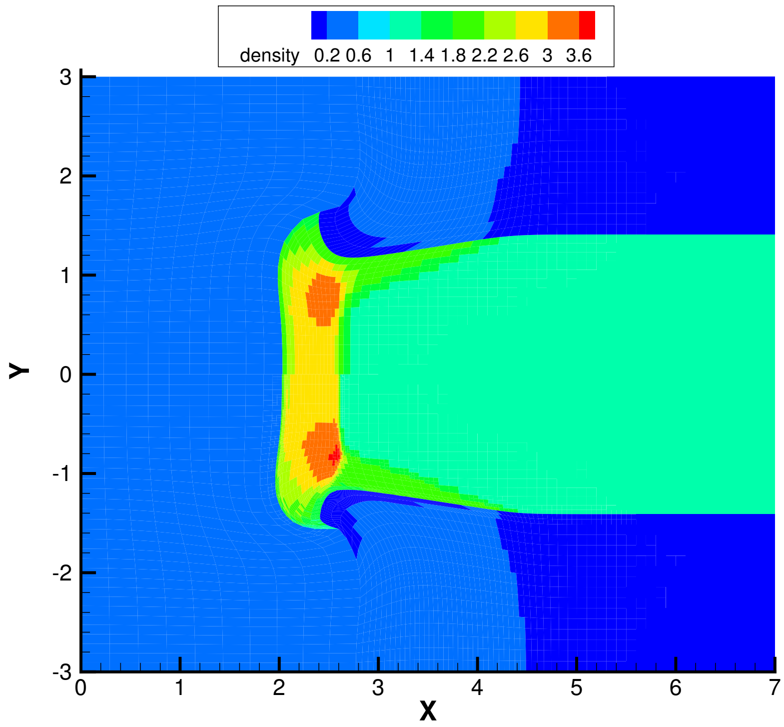

Figure 15: Comparison of the density fields for the triple point problem at $t = 2.0$: base mesh (top) and dynamically adapted AMR mesh (bottom).

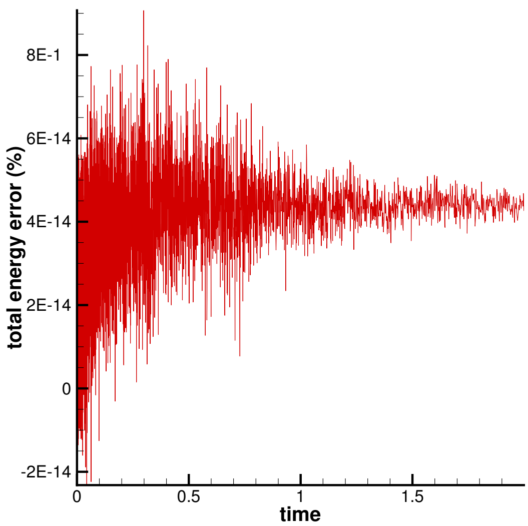

Figure 16: Time history of the total energy relative error for the triple point problem under AMR.